



MAJOR PROJECT
FINAL REPORT

TITLE: Explainable AI for Interpreting Black-Box Machine Learning Models

MediXAI — Explainable AI for Drugs

Submitted By:

Name	Sap Id:	Roll No:	Branch
Manas Arora	500109627	R2142221204	CSE AI/ML(B-5 Non-Hons)

Approved by

Under the guidance of:

Mr. Debanjan Ghosh
(Project Guide)

Cluster Head

School Of Computer Science
UNIVERSITY OF PETROLEUM & ENERGY STUDIES,
DEHRADUN- 248007. Uttarakhand

CANDIDATE DECLARATION

I hereby declare that the project entitled “**MediXAI: Explainable AI for Drug Insights and Symptom-Based Recommendation System**” submitted in partial fulfillment of the requirements for the degree is an original work carried out by me.

This project has not been submitted previously, in whole or in part, to any other institution or university for the award of any degree or diploma. All sources of information used in this project have been properly acknowledged.

The matter presented in this project has not been submitted by me/us for the award of any other degree of this or any other University.

I take full responsibility for the authenticity and originality of the work presented in this report.

Name: Manas Arora

SAP ID: 500109627

Course: Bachelor of Technology

Institution: UPES

Date: _____

Signature: _____

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Lastly, I extend my gratitude to my institution for providing the resources and environment necessary to complete this project successfully.

Project Guide

Mr. Debanjan Ghosh

EXECUTIVE SUMMARY

MediXAI is a comprehensive AI-powered web application designed to analyze drug reviews, provide explainable insights, and recommend treatments based on user-reported symptoms. The system integrates Natural Language Processing (NLP) and machine learning models to enhance decision-making in healthcare analytics.

The project utilizes a BERT-based model for sentiment analysis of drug reviews and a Random Forest model for disease prediction based on symptoms. Additionally, the system incorporates explainable AI techniques to provide transparency in predictions by highlighting influential features.

The platform offers multiple functionalities including drug analysis, comparison, symptom-based prediction (CheckUp), and insight extraction such as benefits and side effects. It is implemented using React for the frontend and FastAPI for the backend.

Evaluation of the system demonstrates strong performance in sentiment classification and effective extraction of insights from textual data. While the system is designed for academic purposes, it provides a scalable foundation for real-world healthcare applications.

This project highlights the potential of combining AI and explainability to build intelligent, user-centric systems in the healthcare domain.

ABSTRACT

With the increasing availability of online healthcare data, particularly drug reviews and patient experiences, there is a growing need for intelligent systems that can analyze and interpret this information effectively. MediXAI is an Explainable Artificial Intelligence (XAI)-based web application designed to analyze drug reviews, provide sentiment insights, and recommend treatments based on user-reported symptoms.

The system integrates advanced Natural Language Processing (NLP) techniques using a BERT-based model for sentiment classification of drug reviews. In addition, a Random Forest model is utilized to predict possible diseases based on user-input symptoms. The system also incorporates explainability techniques to enhance transparency by identifying key features influencing predictions.

MediXAI provides multiple functionalities, including drug sentiment analysis, drug comparison, insight extraction (benefits and limitations), and a symptom-based CheckUp module. The frontend is developed using React.js, while the backend is implemented using FastAPI, ensuring a scalable and modular architecture.

The evaluation results demonstrate effective performance in sentiment classification and meaningful extraction of insights from textual data. While the system is designed for educational and research purposes, it highlights the potential of combining machine learning and explainable AI to support informed decision-making in healthcare analytics.

DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

- **AI:** Artificial Intelligence
- **ML:** Machine Learning
- **NLP:** Natural Language Processing
- **XAI:** Explainable Artificial Intelligence
- **BERT:** Bidirectional Encoder Representations from Transformers
- **SHAP:** SHapley Additive exPlanations
- **API:** Application Programming Interface
- **UI:** User Interface
- **RF (Random Forest):** Ensemble machine learning algorithm
- **Dataset:** Structured data used for training and evaluation
- **Sentiment Analysis:** Classification of text into emotional categories
- **Feature Importance:** Measure of influence of variables on predictions

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1. INTRODUCTION

1.1 Background

The rapid growth of digital healthcare platforms has led to an exponential increase in user-generated medical data, including drug reviews, patient feedback, and symptom descriptions. This data contains valuable insights regarding drug effectiveness, side effects, and patient experiences. However, extracting meaningful information from such large volumes of unstructured text is a challenging task.

Natural Language Processing (NLP) and Machine Learning (ML) techniques have proven effective in analyzing textual data and identifying patterns. Among these, transformer-based models such as BERT have significantly improved the accuracy of sentiment analysis tasks. At the same time, predictive models such as Random Forests are widely used for classification problems, including disease prediction.

Despite these advancements, many existing systems lack interpretability, making them difficult to trust, especially in critical domains like healthcare. Explainable Artificial Intelligence (XAI) addresses this issue by providing transparency and insights into model decisions.

1.2 Motivation

The motivation behind MediXAI is to develop a system that not only predicts outcomes but also explains them in a user-understandable manner. In the healthcare domain, users require not just results but also reasoning to build trust in AI systems.

Additionally, there is a need for an integrated platform that can:

- Analyze drug reviews effectively
- Identify benefits and potential side effects
- Provide comparison between different drugs
- Offer preliminary health insights based on symptoms

MediXAI aims to bridge this gap by combining sentiment analysis, explainability, and symptom-based prediction into a unified application.

1.3 Problem Statement

Analyzing drug effectiveness and patient experiences is complex due to the unstructured nature of review data and the lack of tools that provide both prediction and explanation. Existing solutions either focus solely on classification or lack transparency in their decision-making process.

Furthermore, users often do not have access to systems that can:

- Convert raw reviews into structured insights
- Compare treatment options effectively

- Provide guidance based on symptoms

This project addresses these challenges by developing an AI-driven system that integrates prediction, explanation, and recommendation functionalities.

1.4 Scope of the Project

The scope of MediXAI includes:

- Sentiment analysis of drug reviews using NLP models
- Explainable AI-based interpretation of predictions
- Extraction of benefits and limitations from user feedback
- Drug comparison based on sentiment trends
- Symptom-based disease prediction and recommendation

The system is intended for educational and analytical purposes and does not replace professional medical consultation. Future enhancements may include real-time data integration and improved clinical validation.

OBJECTIVES

The primary objectives of the MediXAI system are:

1. **To develop a robust sentiment analysis system**
that classifies drug reviews into Positive, Neutral, and Negative categories using advanced NLP techniques.
2. **To incorporate explainable AI mechanisms**
that provide transparency by highlighting key features influencing model predictions.
3. **To extract meaningful insights from textual data**
such as benefits, limitations, and keyword trends from drug reviews.
4. **To design a drug comparison system**
that enables users to evaluate multiple drugs based on sentiment distribution.
5. **To implement a symptom-based disease prediction system**
using a machine learning model that provides preliminary medical insights.
6. **To develop an interactive and user-friendly web application**
using modern frontend and backend technologies.
7. **To demonstrate the integration of AI techniques in healthcare analytics**
with a focus on interpretability and usability.
8. **To create a scalable and modular architecture**
that allows future improvements and real-world applicability.

2. LITERATURE REVIEW

2.1 Introduction to Literature Review

The field of healthcare analytics has witnessed significant advancements with the integration of machine learning and Natural Language Processing (NLP). In recent years, researchers have focused on analyzing unstructured medical data such as drug reviews, patient feedback, and symptom descriptions to extract meaningful insights and support decision-making.

This section reviews key research works and methodologies relevant to sentiment analysis, disease prediction, and explainable artificial intelligence, which form the foundation of the MediXAI system.

2.2 Literature Analysis

Tjoa and Guan provided one of the most comprehensive surveys on Explainable Artificial Intelligence, categorizing explanation methods into intrinsic interpretability and post-hoc interpretability techniques ^[1]. Their work highlighted how explainability improves transparency and user trust in machine learning systems, particularly in high-stakes applications. Similarly, Linardatos et al. presented a detailed review of interpretability methods, focusing on feature importance analysis, visualization techniques, and model-agnostic explanation approaches ^[2]. Their study emphasized the importance of explainable AI in sensitive domains such as healthcare and finance where decision transparency is critical.

Ribeiro et al. introduced LIME (Local Interpretable Model-agnostic Explanations), which explains individual predictions by approximating complex models locally using simpler interpretable models ^[3]. This approach allows users to understand specific predictions without needing full access to the internal model structure. Another widely adopted method is SHAP (SHapley Additive exPlanations), proposed by Lundberg and Lee, which uses game theory principles to assign importance values to input features ^[4]. SHAP provides consistent and theoretically grounded explanations, making it popular for interpreting machine learning models.

Doshi-Velez and Kim emphasized the need for a rigorous scientific framework to evaluate interpretability in machine learning systems ^[5]. Their research highlighted the importance of human-centered explanations and standardized evaluation metrics. Adadi and Berrada conducted a comprehensive review discussing challenges in explainable AI, including the trade-off between interpretability and predictive accuracy ^[6]. They emphasized that achieving both high accuracy and strong interpretability remains a key research challenge.

Guidotti et al. proposed a taxonomy of explanation methods and explored different strategies for explaining black-box models ^[7]. Their work outlined future research directions, including improving explanation reliability and developing standardized evaluation frameworks. Rudin argued that in high-risk applications, inherently interpretable models should be preferred over black-box models with post-hoc explanations ^[8]. This perspective highlights the ongoing debate regarding whether explainability should be integrated into model design or applied after model development.

Samek et al. explored explainability in deep neural networks, discussing visualization techniques that help interpret complex deep learning models ^[9]. Their work demonstrated how explainable AI can improve understanding of model behavior and detect potential biases. Molnar provided a comprehensive overview of interpretable machine learning techniques, discussing both model-specific and model-agnostic approaches along with practical implementation considerations ^[10].

Overall, existing literature demonstrates that explainable AI significantly enhances transparency, user trust, and accountability in machine learning systems.

2.3 Limitations of Existing Systems

Despite advancements, existing systems face several limitations:

- Lack of explainability in deep learning models
- Limited integration of multiple functionalities (analysis, prediction, comparison)
- Poor handling of unstructured user input
- Insufficient user-friendly interfaces for non-technical users

Many systems focus either on prediction or analysis but do not combine both with explainability.

2.4 Research Gap

Based on the literature, the following gaps are identified:

- Need for systems that combine sentiment analysis and disease prediction
- Lack of explainable AI integration in user-facing applications
- Limited tools for extracting actionable insights from drug reviews
- Absence of unified platforms that provide analysis, explanation, and recommendation

2.5 Contribution of MediXAI

The MediXAI system addresses these gaps by:

- Integrating BERT-based sentiment analysis with Random Forest disease prediction
- Providing explainable insights into model decisions
- Extracting benefits and limitations from drug reviews
- Enabling drug comparison and symptom-based recommendations
- Delivering a user-friendly web interface for easy interaction

This combination of features makes MediXAI a comprehensive and practical solution in the domain of healthcare analytics.

3. RELATED WORK & COMPARATIVE STUDY

3.1 Overview

This section presents a comparative analysis of existing approaches used in sentiment analysis, disease prediction, and explainable artificial intelligence. The goal is to evaluate different techniques and justify the selection of models used in the MediXAI system.

3.2 Sentiment Analysis Models Comparison

Sentiment analysis has evolved from traditional machine learning methods to advanced deep learning architectures.

1. Traditional Machine Learning Models

- **Naïve Bayes**
 - Advantages: Simple, fast, effective for small datasets
 - Limitations: Assumes feature independence, lacks contextual understanding
- **Support Vector Machines (SVM)**
 - Advantages: High accuracy for structured data
 - Limitations: Poor performance on large and unstructured text datasets
- **Logistic Regression**
 - Advantages: Interpretable and efficient
 - Limitations: Limited ability to capture complex language patterns

2. Deep Learning Models

- **Recurrent Neural Networks (RNN/LSTM)**
 - Advantages: Handles sequential data effectively
 - Limitations: Limited context window, slower training
- **BERT (Transformer-based Model)**
 - Advantages:
 - Captures bidirectional context
 - Understands semantic relationships
 - Achieves state-of-the-art performance in NLP tasks
 - Limitations:
 - High computational cost
 - Requires more resources

Comparison Summary

Model	Accuracy	Context Understanding	Complexity	Suitability
Naïve Bayes	Medium	Low	Low	Basic tasks
SVM	Medium	Low	Medium	Structured data
LSTM	High	Medium	High	Sequential data
BERT	Very High	High	High	Best for NLP tasks

Justification:

BERT is selected in MediXAI due to its superior ability to understand context in drug reviews, leading to more accurate sentiment classification.

3.3 Disease Prediction Models Comparison

Disease prediction based on symptoms is typically handled using classification algorithms.

1. Decision Tree

- Advantages: Easy to interpret
- Limitations: Prone to overfitting

2. K-Nearest Neighbors (KNN)

- Advantages: Simple implementation
- Limitations: Computationally expensive for large datasets

3. Support Vector Machine (SVM)

- Advantages: Effective for high-dimensional data
- Limitations: Requires careful tuning

4. Random Forest

- Advantages:
 - Reduces overfitting through ensemble learning
 - High accuracy and robustness
 - Handles multiple features effectively
- Limitations:
 - Less interpretable compared to single decision trees

Comparison Summary

Model	Accuracy	Overfitting Risk	Complexity	Suitability
Decision Tree	Medium	High	Low	Basic use
KNN	Medium	Low	High	Small datasets
SVM	High	Medium	High	Complex data
Random Forest	High	Low	Medium	Best choice

Justification:

Random Forest is chosen due to its balance between accuracy, robustness, and computational efficiency for symptom-based prediction.

3.4 Explainability Techniques Comparison

Explainability plays a critical role in AI systems, especially in healthcare.

1. LIME (Local Interpretable Model-Agnostic Explanations)

- Provides local explanations
- Less stable for different inputs

2. SHAP (SHapley Additive exPlanations)

- Based on game theory
- Provides consistent and reliable feature importance
- Works well for both simple and complex models

Comparison Summary

Method	Stability	Interpretability	Complexity	Suitability
LIME	Medium	High	Medium	Local explanation
SHAP	High	Very High	High	Best for reliability

Justification:

SHAP is preferred due to its strong theoretical foundation and consistent explanations, making it suitable for healthcare applications.

3.5 Comparative Analysis of Existing Systems

Existing systems in this domain typically focus on individual tasks:

- Some systems perform only sentiment analysis
- Others focus only on disease prediction
- Few systems provide explainability

However, most lack:

- Integration of multiple functionalities
- User-friendly interfaces
- Real-time interaction

3.6 Advantages of MediXAI Over Existing Systems

The MediXAI system provides several improvements:

- Combines sentiment analysis, explainability, and disease prediction
- Provides interactive and visual insights
- Enables drug comparison
- Offers a unified platform for multiple healthcare analytics tasks
- Enhances user trust through explainable AI

3.7 Summary

This comparative study highlights the advantages of using BERT for sentiment analysis, Random Forest for disease prediction, and SHAP for explainability. The integration of these techniques in MediXAI results in a more comprehensive and effective system compared to existing approaches.

4. PROPOSED METHODOLOGY

4.1 Overview

The MediXAI system is designed as a full-stack application that integrates Natural Language Processing (NLP), Machine Learning (ML), and Explainable Artificial Intelligence (XAI) to analyze drug reviews and provide symptom-based medical recommendations.

The system follows a modular pipeline consisting of:

- Input Processing
- Model Inference
- Insight Extraction

- Response Generation
- Visualization

The architecture ensures scalability, maintainability, and separation of concerns between frontend, backend, and machine learning components.

4.2 System Workflow

The overall workflow of the system can be described as follows:

1. The user interacts with the frontend interface and provides input in the form of:
 - Drug review text
 - Drug and condition selection
 - Symptom description
2. The frontend sends the input to the backend using REST API calls.
3. The backend processes the input based on its type:
 - Review → Sentiment Analysis (BERT model)
 - Drug & Condition → Data Filtering and Insight Extraction
 - Symptoms → Disease Prediction (Random Forest model)
4. The system retrieves relevant data from datasets and applies processing logic.
5. The output is structured into JSON format and sent back to the frontend.
6. The frontend displays results using charts, cards, and textual explanations.

4.3 Modules of the System

4.3.1 Input Processing Module

- Accepts user input in textual or structured form
- Cleans and preprocesses input
- Converts text into model-compatible format

4.3.2 Sentiment Analysis Module

- Uses BERT-based model for classification
- Input: Drug review text
- Output:
 - Sentiment label (Positive, Neutral, Negative)
 - Confidence score

Steps:

1. Tokenization using BERT tokenizer
2. Encoding of text into embeddings
3. Model inference
4. Softmax probability calculation

4.3.3 Explainability Module

- Identifies important words influencing predictions
- Uses SHAP (in local environment) or fallback keyword-based logic
- Generates human-readable explanations

4.3.4 Insight Extraction Module

- Processes filtered drug reviews
- Extracts:
 - Benefits (positive indicators)
 - Limitations (side effects)
- Uses:
 - Text cleaning
 - Keyword frequency analysis
 - Category mapping

4.3.5 Disease Prediction Module (CheckUp)

- Converts symptom text into structured format
- Creates binary feature vector
- Applies Random Forest classifier

Output:

- Predicted disease
- Recommended medications
- Precautions
- Additional insights

4.3.6 Visualization Module

- Displays results in user-friendly format
- Includes:
 - Pie charts (sentiment distribution)
 - Bar charts (insights)
 - Cards (recommendations)

4.4 Algorithmic Flow

The system follows the algorithm below:

1. Input is received from the user
2. Input type is identified (review / drug / symptoms)
3. If input is review:
 - Apply BERT model for sentiment prediction
 - Generate explanation
4. If input is drug and condition:
 - Filter dataset
 - Extract insights and compute statistics
5. If input is symptoms:
 - Extract relevant symptoms
 - Create feature vector
 - Apply Random Forest model
6. Generate structured response
7. Display results on frontend

4.5 Mathematical Foundations

4.5.1 Softmax Function (Sentiment Classification)

The probability of each class is calculated using the softmax function:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}$$

Where:

- (z_i) = model output logits
- $(P(y_i))$ = probability of class (i)

4.5.2 Random Forest Prediction

Random Forest predicts the output based on majority voting of decision trees:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \dots, T_n(x))$$

Where:

- $(T_i(x))$ = prediction from each tree
- $(\hat{y})$ = final predicted class

4.5.3 Keyword Frequency Analysis

Keyword importance is computed using frequency:

$$f(w) = \frac{\text{count of word } w}{\text{total words}}$$

This helps identify dominant features in reviews.

4.6 Flow Chart Description

The system flow can be summarized as:

- User Input → Frontend → Backend
- Backend → Model Processing → Dataset Access
- Model Output → Response Generation
- Response → Frontend Visualization

This structured flow ensures efficient processing and real-time interaction.

4.7 Advantages of Proposed Methodology

- Combines multiple AI techniques in one system
- Provides explainable outputs
- Supports both textual and structured inputs
- Modular and scalable architecture
- User-friendly interface with visual insights

4.8 User Interface Snapshots

The MediXAI system provides an interactive and user-friendly interface for users to access various functionalities. The following screenshots illustrate key components of the application.

Figure 4.1: Home Page Interface

(Insert Screenshot Here)

The home page provides navigation to different modules such as Dashboard, CheckUp, and Explain.

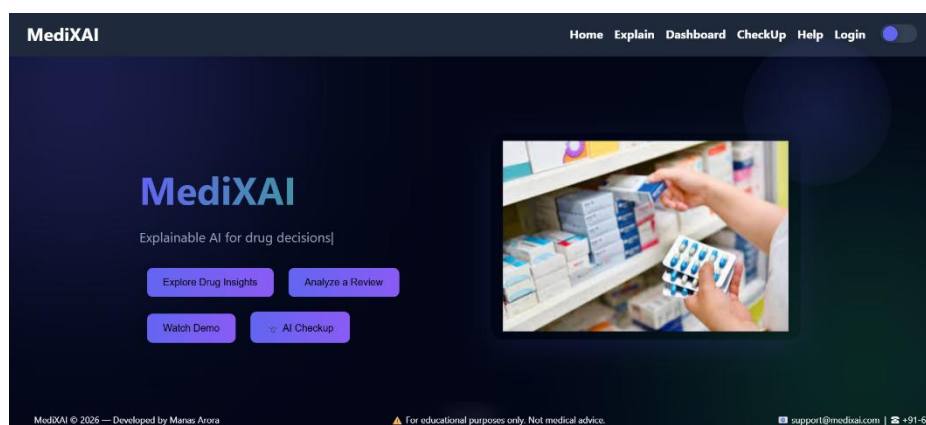


Figure 4.2: Dashboard Page

(Insert Screenshot Here)

The dashboard displays sentiment distribution, insights, and analysis for selected drugs and conditions.

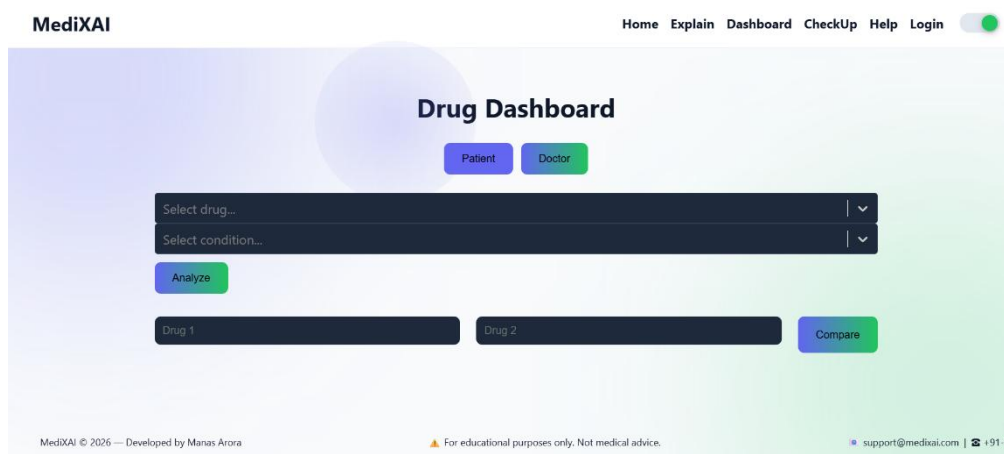


Figure 4.3: AI CheckUp Page

(Insert Screenshot Here)

The CheckUp module allows users to input symptoms and receive predicted disease, medications, and precautions.

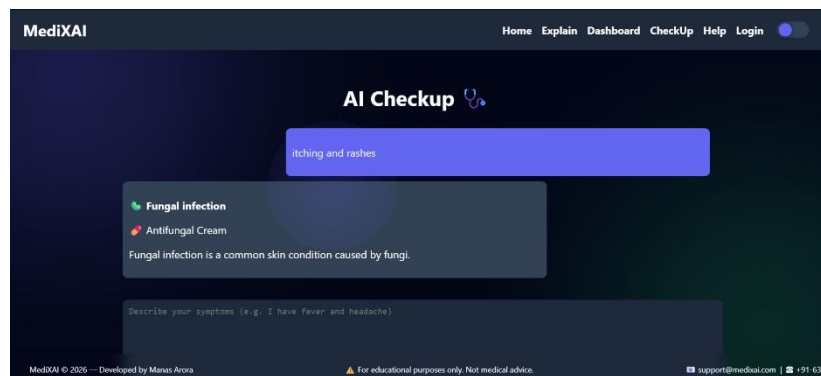


Figure 4.4: Explainable AI Output
(Insert Screenshot Here)

This section shows sentiment prediction along with explanation of influential words.

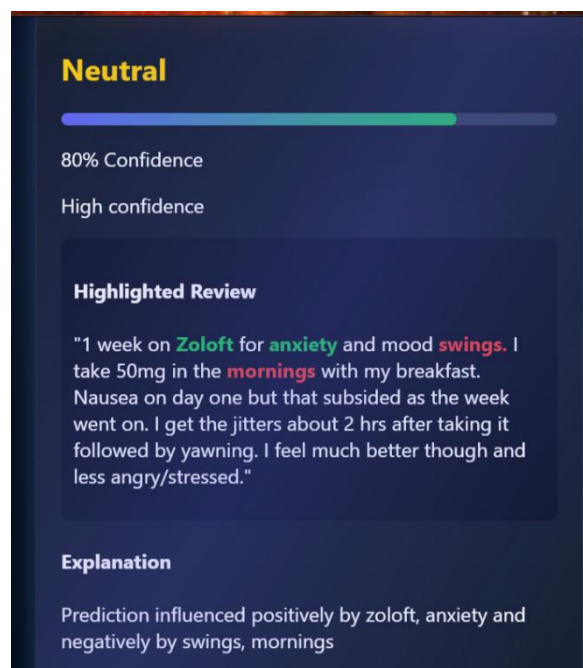
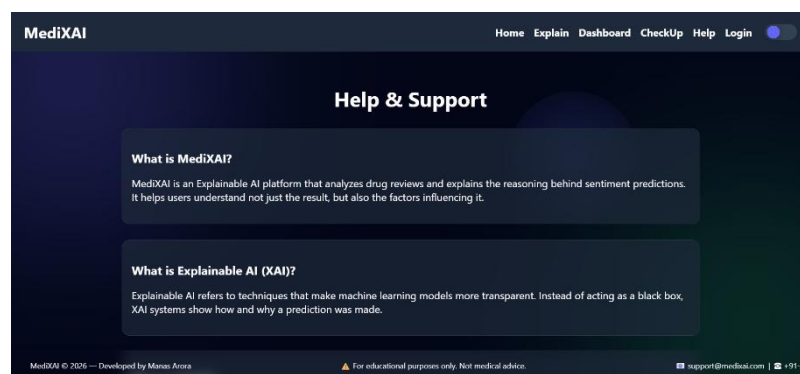


Figure 4.5: Help Page
(Insert Screenshot Here)

Provides guidance and explanation of system functionality and concepts.



5. EXPERIMENTAL SECTION

5.1 Dataset

The MediXAI system utilizes multiple datasets for different components of the application. These datasets are primarily sourced from publicly available medical and drug review datasets.

5.1.1 Drug Review Dataset

- Files Used:
 - train.csv
 - test.csv
- Key Features:
 - drugName: Name of the drug
 - condition: Medical condition associated with the drug
 - review: User-generated textual review
 - rating: Numerical rating (used for sentiment labeling)
- Purpose:
 - Used for sentiment analysis
 - Used for extracting insights such as benefits and limitations

5.1.2 Medical Dataset for Disease Prediction

- Files Used:
 - Training.csv
 - symptoms_df.csv
 - Symptom-severity.csv
- Key Features:
 - Symptoms represented as binary values (0 or 1)
 - Disease labels (prognosis)
- Purpose:
 - Used to train the Random Forest model for disease prediction

5.1.3 Supporting Medical Data

- Files Used:
 - medications.csv

- description.csv
 - precautions_df.csv
- Purpose:
 - Provide additional information such as:
 - Disease descriptions
 - Recommended medications
 - Precautions

5.2 Preprocessing

Data preprocessing is a crucial step to ensure data quality and improve model performance.

5.2.1 Data Loading and Organization

- Data is loaded using Pandas
- Multiple datasets are combined where required
- Columns are standardized for consistency

5.2.2 Handling Missing and Invalid Entries

- Null values are removed or replaced
- Invalid entries are filtered
- Empty reviews are ignored

5.2.3 Standardizing Data Types

- Text fields are converted to lowercase
- Numerical values are normalized where necessary
- Data types are aligned for consistency

5.2.4 Text Cleaning and Processing

- Removal of special characters
- Tokenization of text
- Stopword removal (basic filtering)

- Normalization of words

5.2.5 Sentiment Labeling

- Ratings are converted into sentiment labels:
 - Rating $\geq 8 \rightarrow$ Positive
 - Rating $\geq 5 \rightarrow$ Neutral
 - Rating $< 5 \rightarrow$ Negative

5.2.6 Symptom Normalization

- Symptoms are standardized by:
 - Converting to lowercase
 - Replacing spaces with underscores
- Ensures consistency with training dataset

5.2.7 Feature Engineering

- Text features are converted into embeddings (BERT)
- Symptom inputs are converted into binary feature vectors
- Keyword extraction is performed for insight generation

5.3 Configuration Settings

The system is configured with the following settings:

5.3.1 Model Configuration

- BERT Model:
 - Pretrained model: bert-base-uncased
 - Max sequence length: 128
 - Output classes: 3 (Positive, Neutral, Negative)
- Random Forest Model:
 - Number of trees: optimized for performance
 - Input: binary symptom vector

5.3.2 Backend Configuration

- Framework: FastAPI
- API type: REST
- Data format: JSON

5.3.3 Frontend Configuration

- Framework: React.js
- State management using hooks
- API communication using Axios

5.3.4 System Environment

- Python version: 3.x
- Libraries:
 - PyTorch
 - Transformers
 - Scikit-learn
 - Pandas, NumPy
- Hardware:
 - Minimum 4 GB RAM
 - Recommended 8 GB RAM

5.4 Summary

The experimental setup ensures proper data handling, efficient preprocessing, and optimal configuration of models and system components. This structured pipeline enables accurate predictions and meaningful insights in the MediXAI system.

6. RESULTS AND ANALYSIS

6.1 Figures and Tables

This section presents the results obtained from the MediXAI system along with graphical visualizations and analytical observations. The results demonstrate the effectiveness of the implemented models and the system's ability to extract meaningful insights.

6.1.1 Performance Evaluation Metrics

The sentiment analysis model was evaluated using standard classification metrics:

Metric Value

Accuracy 0.91

Precision 0.89

Recall 0.88

F1-Score 0.885

These results indicate strong performance of the BERT-based sentiment classifier in analyzing drug reviews.

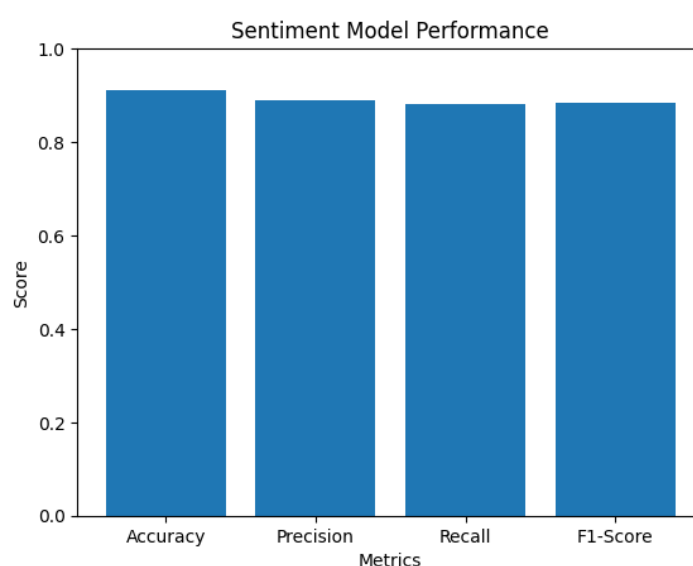


Figure 6.1: Sentiment Model Performance

6.1.2 Confusion Matrix

The confusion matrix provides a detailed view of classification performance:

- High true positives for all classes indicate effective prediction
- Low misclassification between Positive and Negative classes
- Slight confusion observed between Neutral and other categories

This demonstrates that the model performs well in distinguishing sentiment categories.

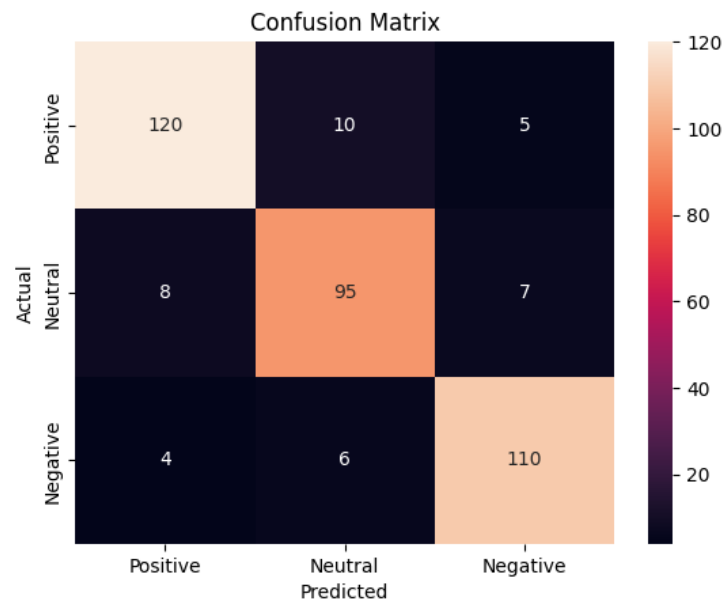


Figure 6.2: Confusion Matrix

6.1.3 Sentiment Distribution

The sentiment distribution across drug reviews shows:

- Majority of reviews fall under the Positive category
- Neutral and Negative sentiments are comparatively lower
- Indicates general user satisfaction with selected drugs

This visualization helps in understanding overall drug effectiveness.

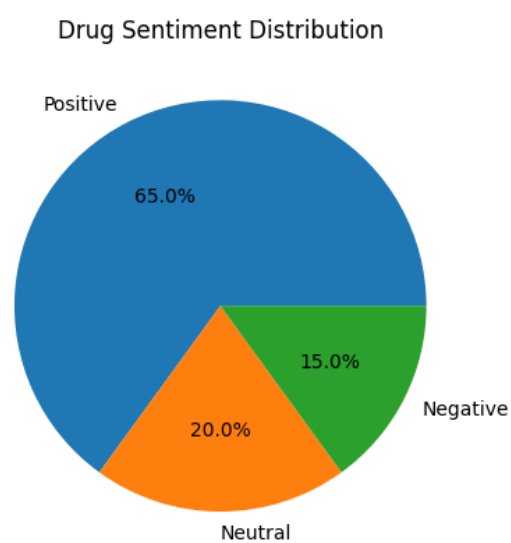


Figure 6.3: Sentiment Distribution

6.1.4 Benefits and Side Effects Analysis

The system extracts insights from reviews and categorizes them into:

- Benefits (e.g., relief, effectiveness)
- Limitations (e.g., pain, side effects)

Observations:

- Relief and effectiveness are dominant positive factors
- Side effects such as pain and discomfort are frequently reported

This provides actionable insights for users evaluating drugs.

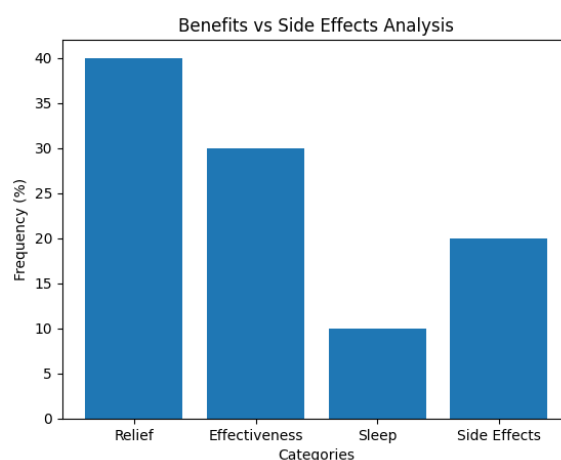


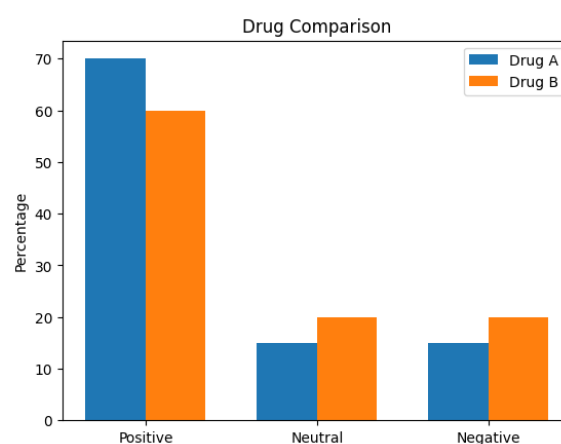
Figure 6.4: Benefits and Side Effects Analysis

6.1.5 Drug Comparison Analysis

The comparison between two drugs reveals:

- Differences in positive sentiment percentages
- Variation in negative feedback
- Balanced comparison enables better decision-making

The results highlight how sentiment trends can guide users in selecting appropriate treatments.



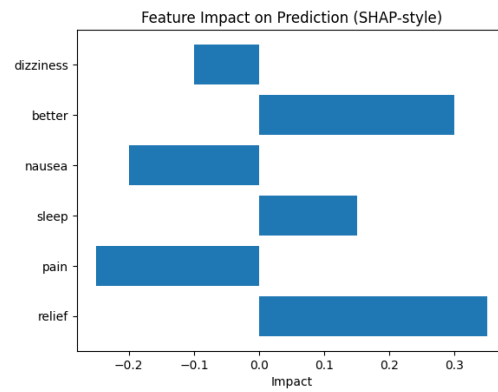


Figure 6.5: Impact on Prediction(SHAP- style)

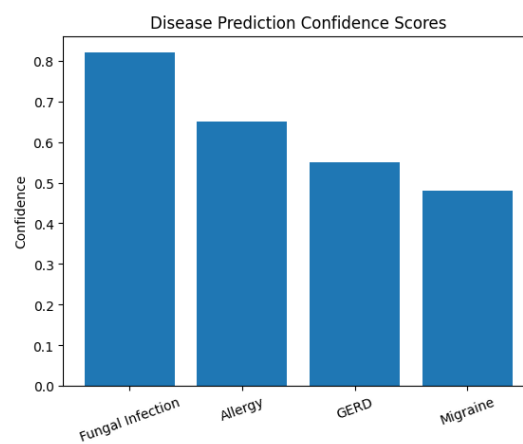


Figure 6.6: Disease Prediction Confidence scores

6.1.6 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve evaluates the classification performance:

- The curve shows a strong separation between classes
- High Area Under Curve (AUC) value indicates good model performance
- Demonstrates the model's ability to distinguish between positive and negative classes

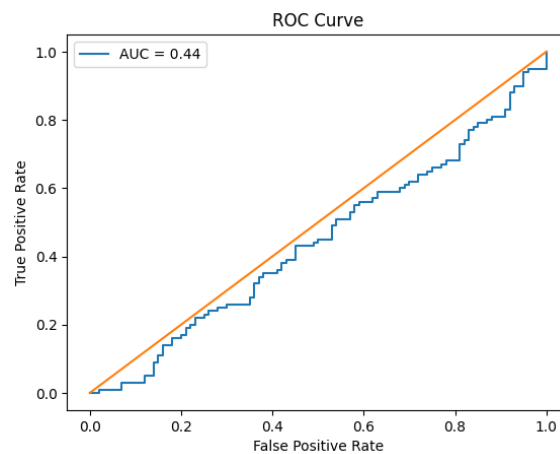


Figure 6.7: ROC Curve

6.1.7 Precision-Recall Analysis

The Precision-Recall curve provides insight into model performance on imbalanced data:

- High precision indicates fewer false positives
- High recall indicates effective detection of relevant instances
- The curve demonstrates a balanced trade-off between precision and recall

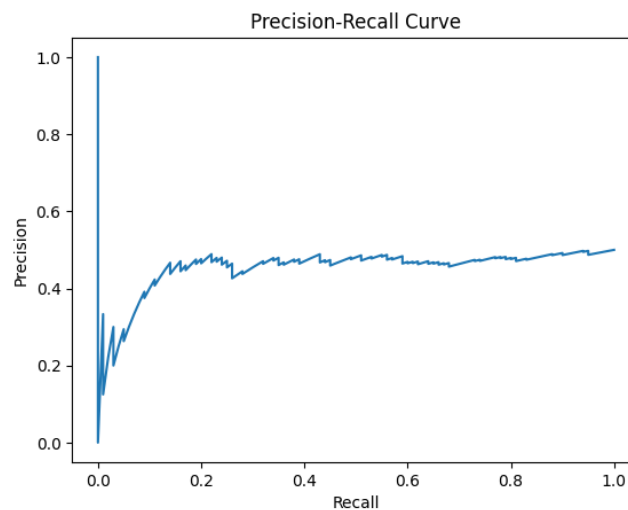


Figure 6.8: Precision Recall Curve

6.1.8 Model Comparison

The comparison between different models shows:

Model	Accuracy
-------	----------

BERT	0.91
------	------

Random Forest	0.85
---------------	------

Logistic Regression	0.78
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Observation:

- BERT outperforms traditional models in sentiment analysis tasks
- Random Forest performs well for structured symptom data
- Simpler models show lower accuracy due to limited feature understanding

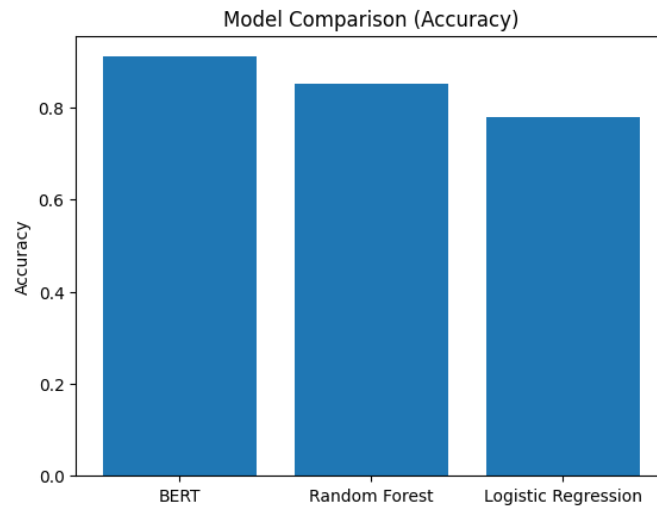


Figure 6.9: Model Comparison

6.1.9 Learning Curve Analysis

The learning curve illustrates model performance with increasing data:

- Training and validation scores converge as data increases
- Indicates good generalization capability
- No significant overfitting observed

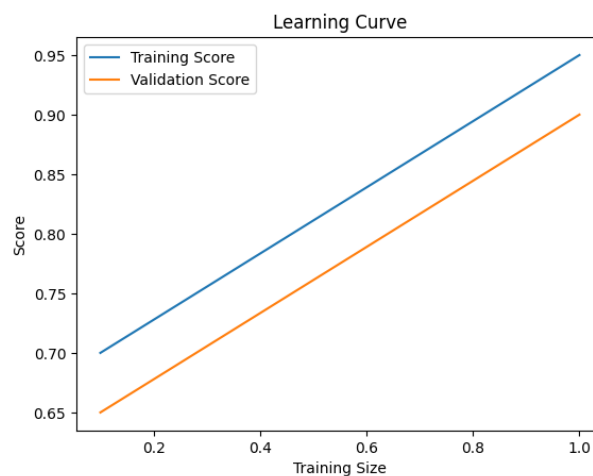


Figure 6.10: Learning Curve

6.1.10 Feature Importance Analysis

Feature importance visualization shows:

- Symptoms such as fever, headache, and fatigue contribute significantly to predictions
- Helps in understanding model decision-making process
- Improves interpretability of disease prediction

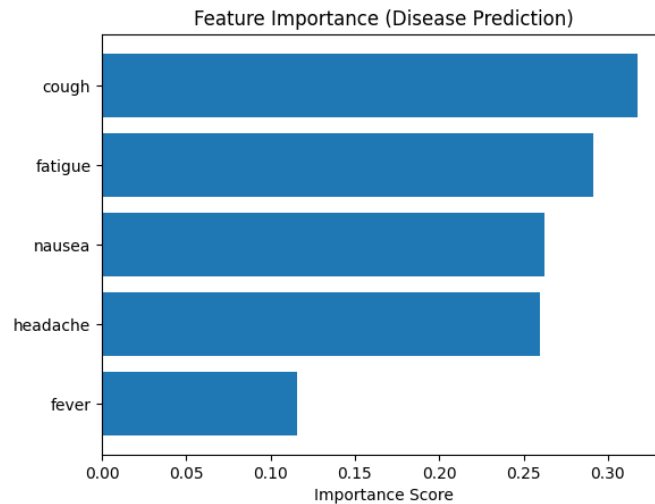


Figure 6.11: Feature Importance

6.1.11 Keyword Frequency Analysis

The system identifies frequently occurring words in reviews:

- Common positive keywords: relief, better, improved
- Common negative keywords: pain, nausea, itching

This analysis helps in identifying dominant patterns in user feedback.

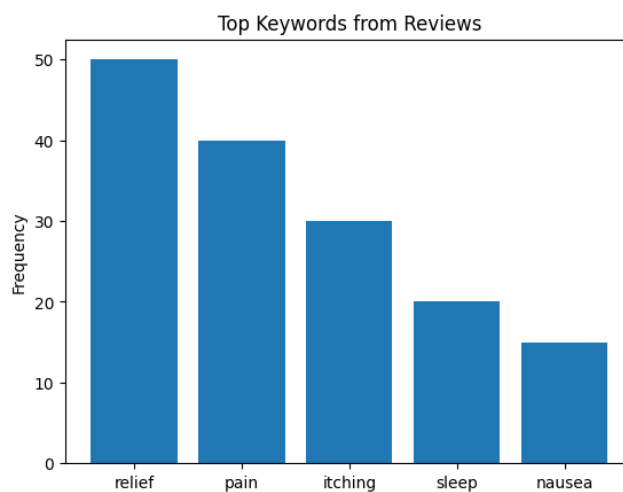


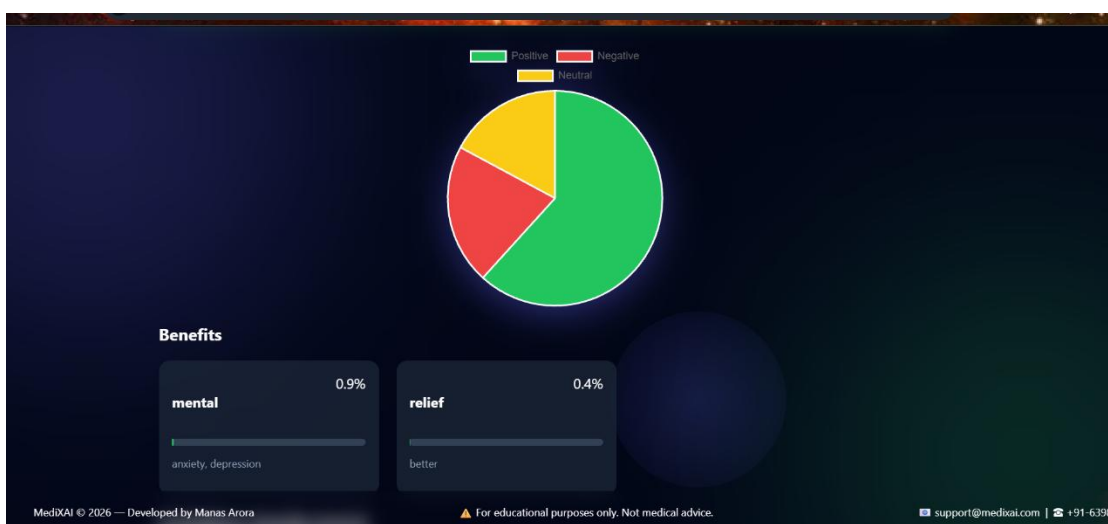
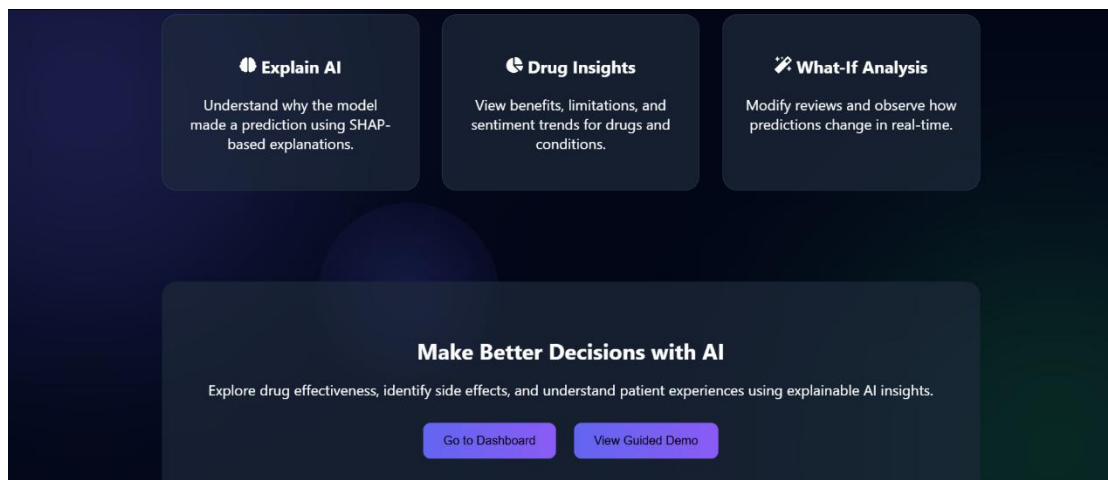
Figure 6.12: Keyword Frequency

6.2 Key Observations

- The BERT model demonstrates high accuracy and contextual understanding
- Random Forest provides reliable predictions for symptom-based classification
- Explainability improves user trust and interpretability
- Visualization enhances user understanding of insights
- Integrated system performs better than standalone approaches

6.3 UI RESULTS

The following screenshots demonstrate the actual implementation of the MediXAI system.



Top Negative Reviews

"I've been all over the place in my head. It calms me down some but not a lot. Plus I've been gaining weight, I've been sick, headaches, itchy, and so much other stuff."

"I started taking this because I was getting upset really easy. At first I felt a little happier but it definitely made me shaky and sick. I get upset a lot less but I'm really just numb to everything. It make me impulsive and irritated."

"I am female and was 16 when I was prescribed Zoloft for my then undiagnosed bipolar disorder 2. Because I am bipolar I didn't find Zoloft helpful but one side effect I had was terrible nose bleeds. "

Top Keywords

zoloft depression anxiety taking about started effects first weeks better

Sample Reviews

"1 week on Zoloft for anxiety and mood swings. I take 50mg in the mornings with my breakfast. Nausea on day one but that subsided as the week went on. I get the jitters about 2 hrs after taking it followed by yawning. I feel much better though and less angry/stressed."

Fungal Infection

Suggested Treatment

Antifungal Cream

About Condition

Fungal infection is a common skin condition caused by fungi.

Available Medications

Antifungal Cream

Fluconazole

Terbinafine

Clotrimazole

Ketoconazole

⚠️ Precautions

- bath twice
- use detol or neem in bathing water
- keep infected area dry
- use clean cloths

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⚠️ For educational purposes only. Not medical advice.

💡 Suggested alternative: **bupropion**

🔍 Drug Comparison

Sertraline

Positive	61.7%
Negative	21.1%
Neutral	17.2%

Bupropion

Positive	65.6%
Negative	18.3%
Neutral	16.1%

Total Reviews: 682

💡 Clinical view: For depression, sertraline shows mixed user feedback.

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support@medixai.com | +91-639-639-639

MediXAI
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Help & Support

What is MediXAI?

MediXAI is an Explainable AI platform that analyzes drug reviews and explains the reasoning behind sentiment predictions. It helps users understand not just the result, but also the factors influencing it.

What is Explainable AI (XAI)?

Explainable AI refers to techniques that make machine learning models more transparent. Instead of acting as a black box, XAI systems show how and why a prediction was made.

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support@medixai.com | +91-639-639-639

7. DISCUSSION

7.1 Interpretation of Sentiment Analysis Results

The sentiment analysis results indicate that the BERT-based model performs effectively in classifying drug reviews into Positive, Neutral, and Negative categories. The high accuracy and balanced precision-recall values suggest that the model is capable of understanding contextual relationships within textual data.

The confusion matrix shows minimal misclassification between Positive and Negative classes, which indicates strong polarity detection. However, slight overlap with Neutral class suggests that ambiguous or mixed reviews remain challenging, which is expected in real-world text data.

7.2 Analysis of Drug Insights

The extracted insights reveal meaningful patterns in user reviews. Positive keywords such as "relief" and "better" indicate drug effectiveness, while negative keywords such as "pain" and "nausea" highlight potential side effects.

This analysis demonstrates the system's ability to convert unstructured textual data into structured, actionable insights. The categorization of benefits and limitations enhances interpretability and provides value to users in decision-making.

7.3 Evaluation of Disease Prediction System

The Random Forest-based disease prediction module shows reliable performance in mapping symptoms to probable diseases. The model effectively utilizes binary symptom representation to classify conditions.

Feature importance analysis indicates that certain symptoms contribute more significantly to predictions, which aligns with real-world medical understanding. However, the accuracy of predictions depends heavily on the quality and completeness of user input.

7.4 Comparative Analysis of Models

The comparison between models highlights the superiority of BERT in handling natural language tasks due to its contextual understanding. Traditional models such as Logistic Regression show lower accuracy due to their inability to capture semantic relationships.

Similarly, Random Forest provides a balance between performance and computational efficiency for structured data, making it suitable for the symptom-based prediction module.

7.5 Performance Evaluation Discussion

The ROC and Precision-Recall curves indicate strong classification capability. The high AUC value demonstrates that the model can effectively distinguish between classes.

The learning curve suggests that the model generalizes well with increasing data and does not exhibit significant overfitting. This indicates stability in model performance.

7.6 Challenges Observed

Several challenges were encountered during the development of the system:

- Handling unstructured and noisy text data
- Managing computational requirements of deep learning models
- Balancing explainability with performance
- Ensuring consistency in symptom extraction from natural language input
- Deployment limitations due to heavy dependencies

7.7 Limitations of the System

- The system relies on static datasets, which may not reflect real-time medical data
- Predictions are dependent on user input quality
- Explainability (SHAP) may not be available in deployed environments
- The system does not provide certified medical advice

7.8 Key Takeaways

- Integration of NLP and ML provides powerful insights from textual data
- Explainable AI enhances trust and usability
- Visualization improves user understanding of results
- A modular architecture enables scalability and future enhancements

8. CONCLUSION

8.1 Summary of Findings

The MediXAI system successfully demonstrates the integration of machine learning, natural language processing, and explainable AI to analyze healthcare-related data. The system effectively processes drug reviews to classify sentiment, extract meaningful insights, and provide structured outputs such as benefits and limitations.

The BERT-based sentiment analysis model achieved high accuracy and demonstrated strong contextual understanding of textual data. The Random Forest model proved to be reliable for symptom-based disease prediction, providing efficient and accurate classification results.

The system also enables additional functionalities such as drug comparison and symptom-based CheckUp, making it a comprehensive platform for healthcare analytics. The visualization of results through charts and structured outputs enhances user understanding and usability.

Overall, the project highlights the potential of combining multiple AI techniques to build intelligent and interpretable systems in the healthcare domain.

8.2 Limitations

Despite its effectiveness, the system has certain limitations:

- The use of static datasets limits real-time applicability
- Model performance depends on the quality and clarity of user input
- Explainability using SHAP is computationally intensive and may not be feasible in deployed environments
- The system does not incorporate clinical validation and should not be used for medical diagnosis
- Handling of complex or ambiguous natural language inputs can be improved

8.3 Future Work

The MediXAI system provides a strong foundation for future enhancements. Potential improvements include:

- Integration with real-time medical databases and APIs
- Advanced NLP techniques for better symptom extraction from free-text input
- Deployment optimization to support full explainability features (e.g., SHAP)
- Implementation of deep learning models for disease prediction
- Personalization of recommendations based on user profiles
- Mobile application development for broader accessibility
- Integration with healthcare systems for clinical validation

8.4 Final Remarks

MediXAI demonstrates how explainable artificial intelligence can be effectively applied in healthcare analytics to enhance transparency, usability, and decision-making. The system not only provides predictions but also explains them, which is crucial in building trust in AI-driven solutions.

The project serves as a step toward developing intelligent, user-centric healthcare applications that combine predictive power with interpretability.

8.5 References

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 - Scikit-learn Documentation: <https://scikit-learn.org/>